CISLUNAR WATCH

Cislunar Space Situational Awareness

Intelligence & Analytics for the New Space Domain

Preliminary Business Plan · March 2026

Vision

Become the definitive intelligence layer for cislunar space situational awareness — the domain nobody is watching yet, that everyone will need within five years.

Why This Works

Starts with free public data and a laptop. No hardware required to begin. Cislunar is underserved by incumbents. The market is forming right now.

1. Executive Summary

CislunarWatch is a space situational awareness (SSA) analytics company focused exclusively on the cislunar domain — the volume of space between Earth and the Moon, including lunar orbit. We aggregate publicly available and licensed tracking data, apply advanced orbital mechanics modeling, and deliver actionable intelligence products to satellite operators, government agencies, launch providers, insurance underwriters, and the growing community of lunar mission operators.

THE CORE INSIGHT

Every existing SSA company is built around Low Earth Orbit. The cislunar domain — where all lunar activity happens — is functionally unwatched. As Artemis missions multiply, commercial landers proliferate, and lunar orbital stations are built, the collision risk and traffic management problem will become acute. CislunarWatch becomes the expert before the market is crowded, not after.

Metric	Detail

Company Name	CislunarWatch (working name)
Business Model	Data analytics + intelligence products as a service
Starting Point	Free public data (Space-Track.org) + open source orbital mechanics libraries
First Customer Target	NASA Artemis mission planners, commercial lunar lander operators
Market	$11.4B+ lunar exploration tech; $1.2B+ commercial SSA market growing rapidly
Capital to Start	Near zero — laptop, free data, Python. Serious product: $500K-$2M seed
Unfair Advantage	First mover in underserved domain; expertise built before competition arrives

2. The Problem

Orbital Space Is Getting Crowded — Especially Near the Moon

Low Earth Orbit has a well-documented debris and congestion problem. But the cislunar domain — everything beyond GEO out to lunar distance and including lunar orbit — is largely unmonitored commercially, and activity there is about to increase dramatically.

- NASA Artemis program will put humans at the lunar south pole. Multiple missions, multiple vehicles, orbital rendezvous, Gateway station in NRHO orbit.
- Commercial landers: Intuitive Machines, Astrobotic, Firefly, ispace — multiple missions per year planned through 2030.
- China's lunar program: crewed south pole mission before 2030, independent orbital infrastructure.
- SpaceX Starship lunar cargo deliveries targeted 2027-2028 — large vehicles in cislunar space on a recurring basis.
- Proposed lunar orbital stations, communication relay satellites, navigation beacons — all adding objects to track.

THE GAP

The US Space Force's 18th Space Control Squadron tracks objects in LEO and GEO. Cislunar space has no equivalent monitoring infrastructure. The orbital mechanics are genuinely more complex — three-body problem dynamics, lunar gravity perturbations, chaotic trajectories near Lagrange points. Existing LEO tools don't transfer cleanly. Nobody has built cislunar SSA as a dedicated commercial product.

Why This Matters

A collision in cislunar space or lunar orbit is categorically different from a LEO collision. There is no atmospheric drag to clean up debris over time. Objects in lunar orbit or at Lagrange points can remain there for centuries. A fragmentation event near the Moon could contaminate the entire lunar orbital environment — permanently degrading humanity's ability to operate there. The stakes are civilizational, not just commercial.

3. The Opportunity

A Domain Nobody Owns Yet

The commercial SSA market is real and growing. LeoLabs, ExoAnalytic, and AGI (with STK software) are established players. But they are all fundamentally LEO-oriented. Their sensor networks, their orbital models, their customer relationships — all optimized for the LEO environment.

Cislunar SSA has different requirements:

- Different orbital mechanics: three-body dynamics, lunar gravity harmonics, halo orbits, libration point stability — none of this is well served by LEO propagators.
- Different sensor geometry: ground-based radar that covers LEO doesn't see deep cislunar space. Different telescope networks, different tracking cadences required.
- Different customers: lunar mission operators, Gateway program, Artemis mission planners — not the LEO satellite constellation operators that LeoLabs serves.
- Different regulatory context: no established cislunar traffic management framework exists yet — whoever builds the intelligence layer helps shape the standards.

> **TIMING**
> *The cislunar market is exactly where LEO was in 2010 — real activity beginning, no established commercial intelligence infrastructure, window open for a first mover to define the space. LeoLabs was founded in 2016 and is now worth hundreds of millions. The cislunar equivalent company founded in 2026 has the same trajectory available.*

4. Starting With Zero Hardware

The Data Is Already Out There

The critical insight: you do not need to own sensors to start. The data layer and the sensor layer are separable. The intelligence and analytics business can be built first, with proprietary sensing added later once revenue funds it.

Free Data Sources Available Today

Source	What It Gives You
Space-Track.org (free)	Two-Line Element sets for ~27,000 tracked objects; updated multiple times daily; register and access today
NASA JPL Horizons	Precise ephemeris data for all solar system bodies including Moon, Lagrange points, and spacecraft; free API
NASA SPICE kernels	Spacecraft trajectory data for past and current missions; free download
ESA DISCOS database	Launch history, object characteristics, fragmentation records; free registration
NASA Orbital Debris Program	Research datasets, size distribution models, collision probability tools; public
Celestrak.org	Curated TLE datasets, supplemental data, free and actively maintained
Space Fence (public products)	US Space Force publishes conjunction data messages (CDMs) for close approaches; free

Licensable Data (Phase 2)

Source	What It Adds
LeoLabs API	Higher precision radar tracking; reduces position uncertainty significantly
ExoAnalytic	Optical tracking data; GEO and beyond; fills gaps in radar coverage
Commercial imagery satellites	Context data for orbital environment characterization

Open Source Tools

The orbital mechanics software stack is mature, free, and well-documented:

```
# Core Python libraries to install immediately
pip install skyfield          # High-precision astronomy; JPL ephemerides
pip install sgp4              # Standard TLE propagator; satellite
position from TLEs
pip install poliastro         # Orbital mechanics; cislunar trajectory
modeling
pip install astropy           # Astronomy calculations; coordinate
transforms
pip install plotly            # 3D visualization of orbital data
pip install pandas numpy      # Data handling

# Key resources
# space-track.org             - Register free; download full TLE catalog
# ssd.jpl.nasa.gov/horizons   - JPL ephemeris API
# celestrak.org               - Curated TLE sets
```

5. What You Build

Phase 1: The Analytics Layer (Laptop Scale)

The first product is not a SaaS platform. It is a demonstrated capability — analysis, visualization, and published insight that establishes you as the person who understands cislunar SSA before anyone else does.

Week 1 Deliverable: Cislunar Object Catalog

Pull all TLEs from Space-Track. Filter for objects with semi-major axes beyond GEO (~42,000 km). Visualize their current positions and projected trajectories in 3D. You will be looking at a small catalog — perhaps a few dozen active objects, some debris. This is the entire currently-tracked cislunar population. You can hold it in your head. That's the point — you can become the expert on every single object in this domain.

Week 2 Deliverable: Artemis Conjunction Analysis

Pull the planned trajectory data for upcoming Artemis missions from NASA SPICE kernels. Calculate minimum approach distances between Artemis vehicles and every tracked cislunar object over the mission timeline. Identify any conjunctions worth flagging. Write it up. This is the kind of analysis NASA's mission planners do internally — but nobody is publishing it as independent analysis.

Week 3 Deliverable: Publish Something

Post your analysis publicly. A Substack article. A LinkedIn post with visualizations. A GitHub repository with your code and methodology. You are now publicly a person who thinks rigorously about cislunar situational awareness. That is a position nobody else has staked out yet. The right people will find you.

> **WHY PUBLISHING MATTERS**
> *Expertise without visibility is invisible. In a domain this new, public analysis creates the reputation that leads to conversations, which lead to contracts, which lead to a company. Sagan didn't just do science — he made it visible. The published analysis is not marketing. It is the work, made accessible.*

Phase 2: The Intelligence Product

Once the analytical foundation is established and initial conversations with potential customers have validated what they actually need, build the first commercial product:

- Cislunar Conjunction Warning Service: automated daily reports of close approaches between tracked objects and planned mission assets in cislunar space. Delivered as API or email digest.
- Mission Safety Analysis: commissioned conjunction analysis for specific missions — pre-launch trajectory review, on-orbit monitoring. High-value, high-margin consulting product.
- Cislunar Domain Awareness Dashboard: web-based visualization and analytics platform; subscription access; shows current object positions, predicted trajectories, risk assessments.
- Regulatory Intelligence: as cislunar traffic management frameworks develop, operators need to understand the emerging rules. Monitoring and analysis of regulatory developments as a product.

Phase 3: Proprietary Sensing

Once revenue funds it, CislunarWatch invests in proprietary sensor capability — not to replicate existing networks but to fill the specific gap that matters for cislunar: deep space optical telescopes capable of tracking objects at lunar distances. A network of 2-3 small telescopes at dark sky sites, combined with the analytics platform, creates a defensible data moat. This is a $2-5M investment justified by the revenue base built in Phases 1 and 2.

6. Business Model

Revenue Stream	Timeline	Estimated Value

Mission Safety Analysis (consulting)	2026-2027	$25K-$150K per mission analysis; high margin; funds early operations
Data & Intelligence API	2027-2028	$5K-$50K/month per customer; recurring; scales with customer count
Dashboard Subscriptions	2027-2028	$1K-$10K/month per operator; SaaS model
Government Contracts (SBIR/STTR)	2026-2027	$300K-$2M non-dilutive; NASA, Space Force, NRO all fund this domain
Regulatory Compliance Products	2028+	As frameworks develop; potentially large market
Proprietary Data Licensing	2029+	After sensor network is built; highest margin product

> **REVENUE SEQUENCING**
> *Consulting revenue funds operations while the platform is built. Platform subscriptions create recurring revenue that justifies seed investment. Seed investment funds the sensor network. Sensor network creates proprietary data moat that justifies Series A. Each stage funds the next.*

7. Market

Target Customers

Segment	Specific Need
NASA Artemis Program	Independent conjunction analysis for Gateway, landers, crew vehicles in cislunar space; mission safety documentation
Commercial Lunar Operators	Intuitive Machines, Astrobotic, Firefly, ispace — all need orbital safety analysis for their missions
SpaceX Starship Lunar	As Starship begins lunar cargo missions, SpaceX needs cislunar traffic awareness
Space Insurance Underwriters	AXA XL, Marsh, Lloyd's syndicates — need independent risk assessment to price lunar mission policies
US Space Force / NRO	Cislunar domain awareness is a stated national security priority; commercial intelligence products fill gaps
ESA / JAXA / ISRO	International agencies planning lunar operations need independent

	non-US-government data
Academic / Research	Universities modeling cislunar environment; research partnerships build credibility and data access

Competitive Landscape

CislunarWatch's primary competitive advantage is domain focus. Current players:

- LeoLabs: best-in-class LEO radar network; no meaningful cislunar capability; different customer base.
- ExoAnalytic: optical tracking; GEO-focused; some deeper space capability but not cislunar-specialized.
- AGI / Ansys STK: dominant software platform for orbital analysis; sells tools, not intelligence products; potential partner not competitor.
- NASA / Space Force internal: do their own analysis but need independent commercial augmentation; potential customer.

> **THE MOAT**
> *Domain expertise plus proprietary data plus network effects — as more operators subscribe, the platform gets smarter, which attracts more operators. First mover in a domain nobody owns yet is the most defensible position in any market.*

8. Milestones & Timeline

Phase	Period	Milestones
Phase 0: Begin	This week	Space-Track account; pull TLE data; install Python stack; first cislunar object visualization; call NASA cousin
Phase 1: Establish	Weeks 1-8	Publish 3-5 analysis pieces; build cislunar conjunction tool; first conversations with NASA/commercial operators; identify technical co-founder
Phase 2: Validate	Months 3-6	First paying mission analysis contract; SBIR Phase I application; incorporate; begin building advisory network
Phase 3: Product	Months 6-18	SBIR award; MVP dashboard; first SaaS subscribers; seed

		fundraise ($500K-$2M)
Phase 4: Scale	Year 2-3	Series A; first proprietary telescope; full platform; government contracts
Phase 5: Dominance	Year 3-5	Proprietary sensor network; industry standard platform; acquisition or IPO path

9. Funding Strategy

CislunarWatch is unusual in that meaningful progress requires almost no capital to start. This is a significant advantage — you can build real credibility and early revenue before touching outside money.

- Phase 0-1 (now): Zero external capital. Free data, open source tools, time. Output: published analysis, early customer conversations, demonstrated expertise.
- Phase 2 (months 3-6): Consulting revenue self-funds. First mission analysis contracts at $25-150K each cover operations. No dilution.
- SBIR Phase I ($300K): Apply to NASA SBIR and Space Force SBIR. Topic areas: cislunar SSA, space traffic management, orbital conjunction analysis. Non-dilutive.
- SBIR Phase II ($2M): Funds platform build and initial sensor exploration. Still non-dilutive.
- Seed ($500K-$2M): Once platform has initial subscribers and SBIR awarded. Target: Seraphim Capital, Space Capital, In-Q-Tel (given national security angle), DARPA SBIR.
- Series A ($10-20M): Post government contracts and proven SaaS metrics. Funds proprietary sensor network.

KEY INSIGHT

Most space companies need tens of millions before they have anything. CislunarWatch has real products and real customers before it needs significant outside capital. That negotiating position is enormously valuable when you do raise.

10. Action Items

This Week — Non-Negotiable

1. Go to space-track.org. Register for a free account. This takes 10 minutes. Do it today.
2. Install the Python stack: pip install skyfield sgp4 poliastro astropy plotly pandas numpy
3. Pull the full TLE catalog. Filter for cislunar objects (semi-major axis > 42,000 km). Count them. Visualize them. You are now looking at the entire tracked cislunar population.
4. Call your NASA cousin. Tell him what you're building. Ask who he knows in cislunar operations or SSA.

Week 2-3

5. Pull NASA SPICE data for Artemis II trajectory. Calculate conjunctions with tracked objects over mission timeline. This is your first real analysis product.
6. Set up a Substack or GitHub. Publish the analysis. Title it something like: 'Cislunar Space: What's Up There and Who's Watching.' You are now publicly in this domain.
7. Find and read everything published by: Moriba Jah (UT Austin, space sustainability), Marlon Sorge (Aerospace Corp, debris), T.S. Kelso (Celestrak). These are the people whose work you're building on.

Month 1-2

8. Identify 10 potential first customers. Find the mission planning leads at Intuitive Machines, Astrobotic, and Firefly on LinkedIn. Send a short note: you're building cislunar SSA tools and would value 20 minutes of their time to understand their needs.
9. Apply to the Lunar Surface Innovation Consortium (LSIC) — they run working groups on exactly this and it gets you in the room with the right people.
10. Research current NASA SBIR solicitations. Identify the 2-3 topic areas most relevant. Draft a 1-page concept before the solicitation closes.
11. Have the Wharfinger conversation with yourself: what does the path to self-sustaining look like, and what's the earliest you could create a clean transition?

11. Why You Specifically

Most people building in this space come from aerospace or defense. They know the orbital mechanics but they don't know how to build a data business, manage distributed technical systems, or navigate complex multi-stakeholder customer relationships. You have the opposite problem in the best possible way — and it's actually the rarer skill set at this stage.

Your Background	Why It Maps
Physics foundation + engineering school	You can read the orbital mechanics literature, understand the math, have real conversations with the technical people
SaaS platform builder (Wharfinger)	You know how to build a data product, manage an API, think about multi-tenant architecture — the platform is familiar territory
Systems integrator (SIGNITEK/Zoho)	You know how to connect disparate data sources, build workflows, deliver to enterprise customers — this is exactly what SSA analytics requires
Project management (PMP/CSM)	Government and aerospace contracts require disciplined PM; you have the credentials and the practice
NVC + relational depth	Customer discovery in a new market is fundamentally a listening skill; you have unusual capacity here
Contemplative practice	Long time horizons, comfort with uncertainty, ability to stay grounded in a high-stakes uncertain venture — underrated founder qualities

THE REAL REASON

You said staring into space makes you want to cry. That is not a hobby response. That is a vocation response. The most durable companies are built by people for whom the problem is personal and the mission is larger than the business. CislunarWatch is a company that matters — keeping orbital space safe and navigable is infrastructure for the entire human future in space. That is a reason to get up in the morning that inventory management cannot compete with.

12. Your First Week — Technical Walkthrough

Concrete enough that you can open a terminal and begin today:

Step 1: Get the Data

```
# 1. Register at space-track.org (free)
# 2. Download the full catalog via their API:

import requests, json

session = requests.Session()
session.post('https://www.space-track.org/ajaxauth/login',
  data={'identity': 'YOUR_EMAIL', 'password': 'YOUR_PASSWORD'})
```

```
# Get all TLEs
resp = session.get(
  'https://www.space-track.org/basicspacedata/query/class/tle_latest/
   ORDINAL/1/EPOCH/%3Enow-30/orderby/NORAD_CAT_ID/format/json')

tles = json.loads(resp.text)
print(f'Tracking {len(tles)} objects')
```

Step 2: Filter for Cislunar

```
from sgp4.api import Satrec, jday
from skyfield.api import load, EarthSatellite
import datetime

# Cislunar = semi-major axis beyond GEO (~42,164 km)
# Filter TLEs by mean motion (lower = farther out)
# GEO mean motion ~ 1.0027 rev/day

cislunar_objects = []
for tle in tles:
    try:
        mean_motion = float(tle['MEAN_MOTION'])  # revs per day
        if mean_motion < 0.5:  # beyond GEO
            cislunar_objects.append(tle)
    except: pass

print(f'Cislunar objects: {len(cislunar_objects)}')
# You will probably see 20-50 objects. You now know them all.
```

Step 3: Visualize

```
import plotly.graph_objects as go
import numpy as np

# Plot Earth + Moon + all cislunar objects
# Moon distance: ~384,400 km
# Use JPL Horizons via Skyfield for Moon position

from skyfield.api import load
ts = load.timescale()
eph = load('de421.bsp')  # JPL ephemeris
earth, moon = eph['earth'], eph['moon']
```

```
t = ts.now()
moon_pos = (moon - earth).at(t).position.km

# Now plot everything in 3D — Earth at origin,
# Moon at moon_pos, your cislunar objects scattered between.
# This is the full picture of what humanity has put
# in cislunar space. It fits on one screen.
```

That's day one. By the end of the week you have conjunction analysis running. By week three you have something worth publishing. The gap between 'thinking about starting' and 'actually started' is a terminal window and a free account.

Closing Note

Two plans in one day — LunarMark and CislunarWatch. That's not indecision. That's rapid iteration toward the right framing. CislunarWatch is the better starting point: lower capital requirement, faster path to customers, skills that map more directly to what you already know how to build, and a domain that is genuinely underserved right now rather than in five years.

LunarMark is still real. The prospecting vision, the data infrastructure for lunar mining, the Earth-as-nature-preserve long game — none of that disappears. CislunarWatch is the vehicle that gets you into the domain, builds the expertise and relationships, and positions you for everything that follows. The two companies are adjacent, not competing.

But none of that matters until you open the terminal.

> **THIS WEEK'S ONLY JOB**
> *Go to space-track.org. Register. Pull the data. Run the first script. Make the call to your cousin. Those four things convert this from a document you read into a thing you are actually doing. The universe has been waiting a long time for you to point yourself at it. It can wait one more week. But only one.*